



DS413 - Introduction to Statistical Learning

Assigned Date: 16/10/2024

11:59pm, Due Date:10/11/2024

data: <https://drive.google.com/drive/folders/1c4NYGUZCLrIWY-eJydKvEyJp4xOGLka-?usp=sharing>

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1 Theory - Comparison of all the algorithms [20]

Compare all the algorithms in terms of performance, decision boundary, complexity (no of parameters and speed) and hyper parameters. (Short Description or a table). (ChatGPT answer is not allowed. Draw the decision boundaries and write your own answer based on your understanding)

Classifiers

- Logistic Regression
- Linear Classifier Estimated using Least Squares.
- Bayes Classifier
- Support Vector Machine Classifier
- Decision Tree for classification

Regression Modules

- Linear Regression
- Ridge or Lasso Regression
- Decision Tree for Regression

2 Programming - Support Vector Machine [40]

Social Network Ads data Suppose a company is going to launch a new campaign for their new brand of car and want to know which category of people are likely to buy their brand new car so they can have the ads that target those peoples. For this they contacted a social network advertising company which have the data from another similar successful campaign. Now, they want to make a model which helps achieve their goal. The dataset contains 400 entries which contains the userId, gender, age, estimated salary and the purchased history. The matrix of features taken into account are age and estimated salary which are going to predict if the user is going to buy new car or not(1=Yes, 0=No)

1. Plot the 2-d data and visualize, comment whether a linear model can separate the data or not. Split the data into
2. Fit SVM model using a python package.
3. Identify support vectors -data points and indicate on the plot. Compute the confusion matrix to find the test accuracy.
4.
 - i) Apply RBF kernel, Gaussian kernel, Polynomial kernel ii) fit the SVM model and iii) Compute the confusion matrix.
5. Compare the SVM model with and without Kernels.
6. Observe the performance of the SVM with respect to "C" value in soft margin SVM. Plot the accuracy versus C value.

3 Feature Importance Using Decision tree [10]

- From Assignment-1.2 question, for the same dataset find out the 3-most important features using decision tree. You can use inbuilt decision tree library? Compare this output with your important features estimated by your linear classification ?

4 Programming-Decision Tree for Regression [20]

1. Use the Boston Dataset for house cost prediction -
2. Dataset [link](#)
3. Implement Decision tree for regression using Library
4. Change the depth of the tree and plot the original and predicted on a single plot. plot the model fits with different values of depth.
5. Plot test RMSE versus depth of the tree.

5 Programming-Implement PCA from scratch [20]

1. Load your MNIST data and preprocess
2. Implement traditional PCA from scratch and project your data into 2D space and visualize.
3. You can label the datapoints in your projection and observe the clusters
4. Use TSNE tool from Sklearn and plot your data on 2D space. (Explore all the options that are there in tsne tool)
5. Comment on tsne tool and traditional pca, which projection is better in terms of discriminability of samples.